

Komodo Dragon Element G

Version 1

Before creating a complete design to scale. We created a mini version of our prototype. First, we had to cut out the measurements in cardboard and use hot glue to glue these pieces together. We created approximately 1.5 of these boxes to show the design and the connection system. Something that this prototype doesn't contain is how we will be incorporating the medical inserts.



This is one box made out of cardboard at about a $\frac{1}{2}$ scale of the real thing. This box had a slit on top to show where the guillotine door would slide in. Once we crafted this box, we made another half box just so our team could tinker with the connection system.



This image shows how the boxes will be connected with some sort of hinge on the final product. Now that we had a design in mind, we started working on and iterating the real prototype.

Version 2

FRAME ASSEMBLY

1. One of our classmates, Corbin Wolfgang, was able to weld some metal to form frames which he brought in on April 8, 2025. This was greatly appreciated as he didn't charge us for his time.



All three box frames and a close up of the door and the insertion point

2. Next gather the plastics: three floors ($\frac{1}{4}$ "") and six sides ($\frac{1}{8}$ "")



One of the floors

3. Cut out the floors for the three boxes using a jigsaw cutter (dimensions: 36 in x 20 in). There should be three identical $\frac{1}{4}$ " thick pieces of polypropylene.



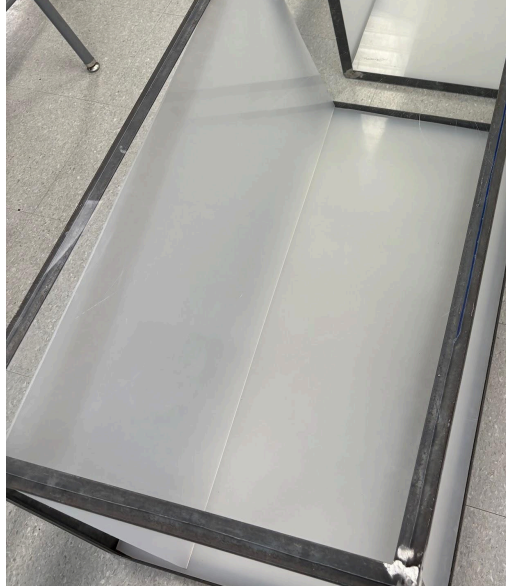
4. Check if they fit in the bottom of the boxes. Trim down if necessary (use an expo marker to draw a line to make sure that the cut is being made in the correct spot)





5. Next choose one box and cut out two identical pieces of $\frac{1}{8}$ " thick polypropylene to use for the side walls which have the same dimensions as the floor (36 in x 20 in)





6. Repeat step four for the remaining two boxes.
7. Next, cut out the ceiling of each of the three boxes using polycarbonate (can't use the jigsaw because the plastic was too thin; use a dremel).
8. Drill bolts into the boxes to keep the polycarbonate and polypropylene in place.
9. Email the Nashville Zoo expert, Nick Hanna to make sure that using Rust-oleum spray paint won't affect the health of the Komodo dragons.



Bennett Lohmeyer Yesterday
To: nhanna@nashvillezoo.org >

Hello Mr. Hanna,
This is the Komodo Dragon Box group.
Would it be alright if we used rustoleum on the box so that we could try to prevent any rust from accumulating on the boxes over time? Would this affect the health of the dragons? Do you have another anti-rust treatment that would work on metal?

Thank you for your input!



Nicholas Hanna Yesterday
To: [Bennett Lohmeyer](#) >

Re: Rust-oleum

Hi Bennett,

Rustoleum will be ok to use on it. They should not be trying to eat or ingest any of it. Thanks for thinking of this!

Nick

10. After we get permission to use the Rust-oleum, spray paint all of the boxes and the doors to give the frame a more appealing look. This spray will also help prevent rusting of the metal frame.



11. Next, drill out holes in the frame for screws for the handles, connection system, and the plastics (walls and floors)
12. Drill the plastics into the frame so that the box is complete. Repeat this step for each of the other two boxes.





With the plan for the frame of the solution complete, our group was also tasked with creating the handles for the Komodo Dragon box. Below are the steps of how we built the handles.

HANDLE MATERIALS

36 8-32 x 1.750" Hex head Screws

22 8-32 Hex Nut

12 8-32 Nylock Nut

12 Hager RC1741 residential door hinges

6 ft T-slot aluminum alloy miter track

12 ft 1" PVC pipe

CONSTRUCTION TOOLS FOR HANDLES

1 11/64" wood drill bit

1 cordless power drill + battery

1 open end wrench

1 dremel

5 .9" cutting discs for dremel

1 .2" hex head drill bit

1 One gallon of liquid flex seal

HANDLES ASSEMBLY

1: Take your 1" PVC pipe and cut it into 6 pieces each 18 inches long

2: Mark out two rectangular holes .5" away from each end with the dimensions of 1.1" X .5"



3: Using the dremel, cut out both holes

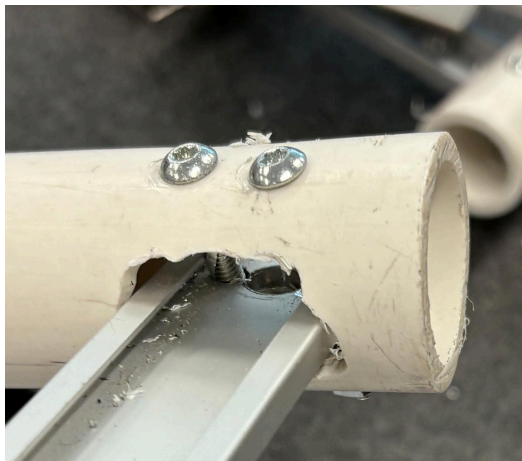
4: repeat step 2 and 3 on the other 5 PVC pieces

5: Take your T-slot aluminum and cut it into 12 pieces each 5.5" long and have the bottom hole

be .2" from the end of the aluminum



6: Slot the opposite end of the aluminum beam from where the hole is .2" from the bottom into the hole you cut in the PVC



7: take the 11/64" drill bit and drill two holes near the end of the PVC pipe going through the aluminum beam and exiting out the other side

8: take two 8-32 x 1.750" Hex head Screws and slot them through the holes you just drilled in step 7

9: take two 8-32 Hex Nut and then tighten them to the end sticking out from the PVC pipe, Repeat this step and steps 7 and 8 on the other end of the PVC pipe

10: repeat steps 7 to 9 on all the other pieces of PVC

11: take two hinges, two 8-32 x 1.750" Hex head Screws and two 8-32 Hex nuts and place the hinge on the flat side of the T-slot aluminum. Make sure that the hinges fold back where the hinge rod is sticking out and the two plates are flat against the T-slot aluminum.

12: Take The bolt and then slide it through the third hole in the hinges and tighten

13: Repeat step 11 and 12 on all other pieces of aluminum

14: Take the dremel and remove the rest of the bolt sticking out from the nut and make it flat

15: Take Black Rustoleum and spray everything making sure to not leave anything uncovered



Final Version

With the handles complete, drill all six of them into designated holes on the boxes.



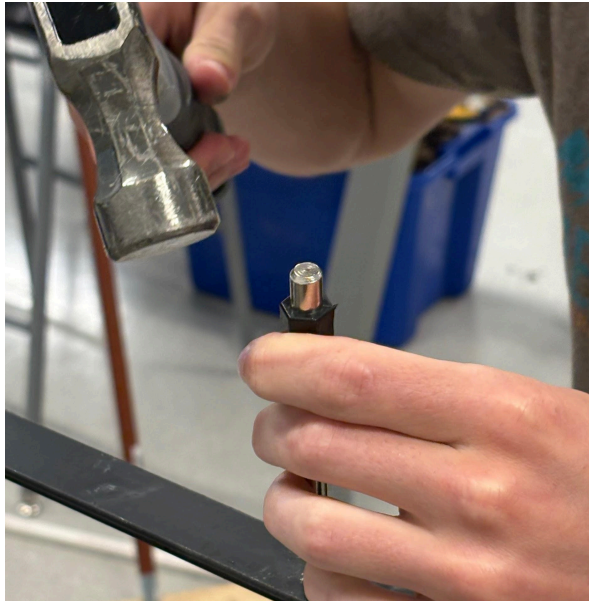
These “H” hole markers marked where the handles should be drilled.



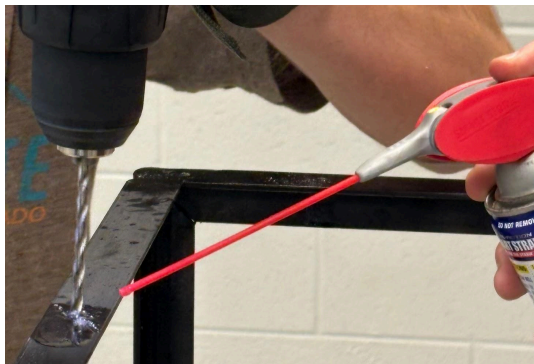
Once the box is complete, work to set up the connection system and medical testing windows with the other two groups working on the Komodo Dragon box. This will complete the final testable prototype.



This is where we marked some other important spots to drill out holes. The “R” is for a railway connection system that one of the other groups designed to help keep the boxes together when they lift the boxes up. The “C” markings are to indicate where the connection system that includes a butterfly hinge will go to prevent the boxes from disconnecting while the Komodo Dragon is inside. Once the dragon is outside the zookeepers can simply disconnect the boxes for easier storage.



This picture shows how we marked the spots where the chalk showed us that we would be drilling. This would make it easier to actually drill the correct spots.





Then, we actually drilled out each whole so we could start using bolts to secure everything to the final prototype. Once the final prototype was complete, we started testing the materials.